PROJECT SELECTION ADDENDUM

GridLock sample validation and MVP plan

Ten supplied examples
Two utility plans

Prioritize GridLock. Demonstrate Jasper–Okatie #2 beside the Goshen–Georgia Pacific rebuild, with visible source dates and incomplete geometry.

Decision	Evidence and limit
Preferred demonstration	Approximately 10.25 miles between screening anchors. Planned milestones: 1 December 2026 and year-only 2028. Shared right-of-way and simultaneous construction remain unverified.
What the audit establishes	All 25 original cross-utility comparisons recalculated; all six workbook proximity results reproduce. Several locations represent the wrong work scope, and schedules require correction.
What is delivered	Validated sample with original values, source dates, coordinate findings, rule ambiguities and the smallest proposed MVP. No project implementation.

Version 1.3.0 addendum • Research cutoff 26 September 2026 • For the four-person ShellHacks team
New user priority supersedes the earlier provisional ReplayBench recommendation; prior study and scores remain dated records.

Challenge requirements and unresolved rules

The supplied Sperry brief requires an interactive comparison of future construction plans from at least two utilities, visual highlighting of overlaps and a ranked list of coordination candidates. A rough cost or impact estimate for one candidate is a bonus. Any technology stack is permitted; public data is required and CEII is excluded. [B, L]

Topic	Source wording or meaning	Provisional implementation decision
Distance boundary	The brief says within 25 miles and closer than 25; the guide says under 25. Both documents use the same numeric threshold.	Use unrounded distance < 25 miles and disclose the strict boundary until clarified. Exactly 25 is excluded under this assumption.
Geography and time	Opening prose allows proximity OR similar timing; detailed instructions make geography primary and time secondary.	Use geographic inclusion for the main list. Do not silently admit distant projects merely because their dates match.
Timing weight	Brief calls time a strong secondary signal; guide calls the day gap a lighter consideration. No numeric weights or timing cutoff.	Show milestone gaps and precision separately. Distance-first ordering is a team choice, not an official scoring formula.
Build window	Brief describes the same build window; guide asks for an in-service-date day gap. These are different measures.	Do not infer simultaneous construction from milestones. Show construction-window overlap as unknown when start/end evidence is absent.

The guide explicitly prescribes the arithmetic midpoint of two located named points, or the one available point, followed by center-to-center distance. That method is consistent within the guide; it is not a requirement for minimum corridor-to-corridor distance. The proxy can miss nearby routes or misrepresent long and partially rebuilt lines.

Clarification needed from Sperry: exact boundary treatment, whether time-only pairs qualify, how timing should affect ranking, and whether a documented proximity candidate with unknown construction overlap satisfies the overlap demonstration. The linked Notion code example was unavailable through public reading; provide its text only if it defines another rule. No alternative numeric radius was found in the supplied DOCX packages.

B = ShellHacks_Challenge_Gridlock.docx, paragraphs 6 and 60–63. L = Finding_Real_Locations_Guide.docx, paragraph 46. Paragraph counts include table cells. Local file modification metadata is August 2026, not a utility publication date.

Dominion sample validation

The original dates match the supplied 2024–2028 plan. Updated IRPs explicitly record three completed projects and the Queensboro–Fort Johnson scope of a fourth. Jasper–Okatie #2 remains a defensible planned-work example: the August 2026 order reports it under construction. [D0–D4]

Sample and identity	Original milestone → refreshed evidence	Coordinate and scope result
DESC_1 6809 E Stevens Creek–Hooks	31 Dec 2024 → in service Aug 2024. D0 p. 14; D1 p. 49 / printed 43.	Stevens Creek point is near a mapped dam substation; Hooks is missing. Single-point proxy only. Do not merge separate 6809 G, now Hooks–Modoc, planned Dec 2027.
DESC_2 6810 A Hooks–Thurmond	31 Dec 2024 → in service Nov 2024. D0 p. 31; D1 p. 49 / printed 43.	Thurmond matches the named 115 kV station; Hooks missing. About 2.3 miles of rebuild, not a point footprint. Historical control.
DESC_3 06367 D–G Jasper–Okatie #2	31 Dec 2025 → planned 1 Dec 2026. Under construction reported 26 Aug 2026. D0 p. 23; D3 p. 12; D4 p. 2.	Use updated Jasper and provisional Okatie station-area anchors; official route map corroborates the named corridor. No surveyed endpoints or route polyline.
DESC_4 6807 B Queensboro compound title	31 Dec 2023 → Queensboro–Fort Johnson in service Dec 2024. D0 p. 1; D1 p. 49 and p. 133.	Queensborough map feature matches the source area; Fort Johnson missing. Bayfront / James Island title components are not fully crosswalked to the completion statement.
DESC_5 6808 S Okatie–Bluffton	1 Jun 2025 → in service May 2025. D0 p. 10; D2 p. 83.	Bluffton matches the named 115 kV station. Okatie point aligns with a 230 kV map feature; exact 115 kV terminal not established. Historical control.

Completion months retain month precision. No first or last day of a month is invented. For DESC_4, the completion evidence applies to the named Fort Johnson rebuild; it does not establish that every component of the compound title is complete.

All locators use one-based PDF pages unless a printed page is separately named. OSM feature IDs, edit dates, raw workbook coordinates and the detailed identity evidence are preserved in validated_sample.json.

Georgia Power sample validation

Every Georgia Power date copied into the workbook's in_service_date column is labeled Need Date in the supplied IRP. Its foreword identifies a December 2024 snapshot. A need date is not confirmed completion, and a passed need date does not establish that work is finished. [G0 p. 4]

Sample and identity	Schedule evidence	Actual work and coordinate result
GPC_1 TEAMS 20793 Evans–Thurmond #5	Need 1 Jun 2033; old start 1 Jun 2029. No newer matching schedule found. G0 p. 410; G1 p. 133.	Only 5.45-mile Euchee Creek–Thurmond section plus Thurmond bus work. Both named workbook stations match OSM, but Evans is outside the rebuild endpoints. Reject the full-line midpoint.
GPC_2 TEAMS 20277 McIntosh reactors	Need 1 Jun 2026; old start 1 Jan 2024. Completion and replacement schedule unresolved. G0 p. 227; G1 p. 77; G2 p. 83.	Reactors at McIntosh plus two 0.1-mile line portions. Coordinate matches named 230 kV McIntosh. Exact equipment and short-line geometry remain unknown.
GPC_3 TEAMS 20065 Goshen rebuild	Need 1 Jun 2027 → latest planned year 2028. Old start 1 Jun 2025 is stale. G0 p. 314; G3 p. 111; G4 p. 53.	6.7-mile Goshen–Georgia Pacific section. Goshen matches OSM. Reject McIntosh as a work endpoint; Georgia Pacific remains unlocated. Use a clearly labeled Goshen-only proxy.
GPC_4 TEAMS 18492 Mitchell–North Tifton	Need 1 May 2025; old start 31 Dec 2021. Actual completion unresolved. G0 p. 233; G1 p. 61.	35.21-mile 230 kV line. Mitchell point matches; North Tifton point maps to a 500 kV yard. Exact 230 kV terminal and route remain unverified.
GPC_5 TEAMS 11821 Jesup–Ludowici	Need 1 Jun 2025; old start 1 Jun 2023. Actual completion unresolved. G0 p. 240; G1 p. 60.	Only 7.5-mile Jesup–North Jesup–Rayonier section. Workbook stations match OSM, but full-line midpoint is outside the validated scope model. Do not merge separate Jesup–Offerman work.

The two workbook "McIntosh" coordinates differ by approximately 0.408 mile. A distinct West McIntosh 500 kV station is nearby. Fixing that longitude would still leave GPC_3's scope wrong. North Tifton's map label likewise does not prove that the broader facility lacks 230 kV equipment; it means the supplied terminal has not been validated.

Latest matching public GPC_3 evidence: preliminary SERTP report dated 12 June 2026. The 22 September 2026 presentation was checked without locating a replacement schedule for these samples. No actual completion date was established for any GPC row.

Defensible demonstration pair and controls

Select DESC_3 and GPC_3 for the demonstration because both have refreshed future milestones and the scope correction is visible. This is a proposed review candidate under the guide's proxy rule. No reviewed sample pair establishes current simultaneous construction, shared right-of-way or savings.

Metric	Original workbook example	Reviewed demonstration
DESC anchor	32.346439, −81.0785475 Arithmetic midpoint of supplied Jasper / Okatie points.	32.34723085, −81.0783793 Midpoint of OSM station-area centers: Jasper way 185380597; provisional Okatie way 1064022697.
GPC anchor	32.3004085, −81.1957885 Full Goshen–McIntosh midpoint.	32.2487012, −81.2094724 Goshen way 1008141064 only. Actual Georgia Pacific work endpoint is missing.
Screening distance	7.55 miles	10.25 miles, rounded from 10.245192665. Mixed midpoint / single-endpoint proxies; not route distance.
Planned milestone gap	517 days from 31 Dec 2025 to 1 Jun 2027.	396–761 days from 1 Dec 2026 to a possible date in calendar 2028. No construction-window overlap inferred.

Original pair	Miles / days	Reviewed disposition
DESC_2 / GPC_1	4.09 / 3074	Historical control: DESC completed; GPC scope and stale schedule require review.
DESC_3 / GPC_2	5.65 / 152	Status unresolved: passed GPC need date, no completion or new schedule found.
DESC_3 / GPC_3	7.55 / 517	Preferred demonstration after the corrections above.
DESC_1 / GPC_1	8.01 / 3074	Historical control: DESC completed.
DESC_5 / GPC_2	14.34 / 365	Historical control: DESC completed; GPC status unresolved.
DESC_5 / GPC_3	14.81 / 730	Historical control: DESC completed despite GPC future plan.

All 25 original pairs were checked: six below 25 miles; the other nineteen are at least 64.77 miles apart. Center caches reconcile, but overlap distances and gaps are hardcoded cells and do not automatically refresh. Calculation uses haversine, radius 3958.7613 miles, on unrounded anchors.

Smallest complete MVP

Build one browser view that lets a planner inspect nearby public projects and see why each deserves follow-up. Use the curated ten-row snapshot as the complete demonstration dataset, with source versions and uncertainty attached. Loading more utilities, live extraction and automated savings estimates are outside this MVP.

Required capability	Smallest complete behavior	Proposed acceptance check
Two utility plans	Load all ten records from a static JSON file. Keep original examples and latest reviewed records separate; identify each utility and work section.	Every visible date and location has a source and precision. Original values remain inspectable.
Interactive map	Pan, zoom and select project anchors. Distinguish utility, completed work, current-plan candidates and unresolved records with text as well as symbols.	Click the preferred pair to see its two proxy types. No straight connector is presented as an actual transmission route.
Ranked candidate list	Cross-utility, future-plan candidates below 25 miles, ordered by unrounded distance; show time gap and geometry confidence. Put stale and completed examples in a separate review view.	Preferred pair appears as a proximity candidate. The 4.09-mile historical pair cannot outrank it as current work. Stable ID tie-break; no invented official weights.
Source inspection	Open a concise evidence panel with work scope, previous and refreshed dates, PDF page links, map feature IDs and unresolved checks.	Switch original / reviewed versions; 7.55 changes to 10.25 miles and 517 days changes to a 396–761-day interval.

A compelling demonstration

In about two minutes, show the six original proximity hits. Select the seemingly attractive 5.65-mile pair and expose its stale Georgia Power need date. Switch to the preferred pair, show the corrected work section and 2028 source, and explain why the candidate is worth reviewing while construction overlap remains unknown. Finish by opening the dated source evidence and listing the exact missing geometry and schedule fields.

Scope and dependencies

Use the team’s familiar web stack, static data and a simple map library subject to current event reuse rules. No live Overpass dependency during judging. Defer OCR, LLM extraction, accounts, extra databases, wallets and sponsor-tool additions. Omit the optional cost bonus until a supported shared work quantity and unit-cost basis exist. If basemap access fails, retain a labeled local coordinate plot and working evidence list.

The planner workflow and traceable corrections are the proposed distinction. Historical study support is limited: 30/31 podium projects documented an interface or demonstrator and 30/31 a clear workflow. These observations support a visible end-to-end demo, not a prediction that GridLock will win.

Four person build plan after a decision to proceed

Assume four software teammates and ordinary laptops; skills and remaining time are unconfirmed. The verified hacking window ends Sunday 27 September at 11:00 EDT, but the submission cutoff may be earlier. Start after a new instruction to implement and reserve two hours before the earlier deadline. [H]

Elapsed work block	Responsibilities	Exit condition
0–1 hour	A: confirm local submission/reuse rules and Sperry ambiguities. B: data rendering. C: map proof. D: demo and independent fixture checks.	Data loads, preferred anchors plot correctly and the proxy-based demo is acceptable. If actual shared construction is mandatory, the sample is insufficient; obtain confirmed routes and windows before making that claim.
1–3 hours	A: freeze curated source records. B: list and evidence panel. C: map selection and status legend. D: comparison calculations and date-precision checks.	One complete preferred-pair flow works, with source dates and unknowns visible.
3–6 hours	B/C: link map and list; add original/reviewed toggle. A/D: review all ten rows, negative controls and accessibility.	All mandatory behavior works. Cut optional features immediately if the complete flow is not ready.
6–8 hours	D leads repeatable acceptance checks; A verifies citations and disclosures; B/C fix defects and package the app.	Rehearsed demo, local backup, no false route or completion claims.
Final 2 hours	All: freeze features, rehearse presentation, record any required video, submit required links and verify receipt.	Submission complete before the actual cutoff; keep a backup copy and time for a failed upload.

Budget 8–12 active team hours, plus a 6–8-hour overnight rest block if the actual start and deadline permit it. Take short breaks. With fewer than eight active hours, cut comparison controls and polish while retaining the map, ranked list and evidence panel. This is an estimate, not confirmed remaining time.

Validation before submission

Reproduce all 25 distances and six matches. Check 24.999, 25.000 and 25.001-mile boundaries, missing coordinates, single-point fallback, year-only date bounds and completed-scope exclusion. Keep distinct IDs and construction phases separate. A teammate independently checks sources and classifications. Proposed performance target: compute all 25 comparisons and update selection within 100 ms on the demo laptop; measure before claiming it.

Source register and unresolved evidence

The riskiest dependency is missing public evidence: Georgia Pacific's endpoint, route geometry and construction windows. Screening can show these gaps, but a verified coordination claim requires resolving them. Check local rules for reuse of this research, prebuilt code and AI disclosure before building. No prior application code is assumed eligible.

The JSON sample preserves all populated workbook cells, formulas, cached values, number formats, original dates, ten project reviews, all 25 calculations, public map IDs and source hashes. It includes detailed official-source locators and unresolved findings. The archive, workbook and supplied documents remain unchanged. Access date for external research: 26 September 2026.

B and L Supplied challenge and location guide; modified metadata 26 Aug 2026. Workbook metadata also 26 Aug 2026. These are file dates, not utility publication dates.

D0 DESC 2024–2028 project descriptions. Publication date unspecified; PDF creation metadata 5 Mar 2024. Open source

D1 DESC 2025 IRP update, filed 31 Mar 2025. Completion table PDF p49; appendix p133. Open source

D2 DESC 2026 IRP, filed 31 Mar 2026. Appendix B, PDF p83. Open source

D3 DESC 2026–2030 descriptions, p12. Publication date unspecified; creation metadata 28 Apr 2026. Open source

D4 PSC Order 2026-482, dated 26 Aug 2026, pp2–5. June 2032 concerns a separate reconductoring package. Open source

G0 Supplied 2025 Georgia Power IRP Volume 3 PUBLIC DISCLOSURE. December 2024 snapshot. Docket link is a landing page; direct official file URL not recovered. Open source

G1 SERTP final plan, 27 Nov 2024; G2 is the 27 Jun 2024 Q2 presentation, p83. Open source

G3 SERTP final plan, 26 Nov 2025, PDF p111 / printed110. Open source

G4 SERTP preliminary plan Non-CEII, 12 Jun 2026, p53. September 22 presentation checked for later updates. Open source

Map Dominion final route selection map, linked as 11 Jul 2022. OSM public snapshots 26 Sep 2026; feature edit dates retained separately. Open source

H Prior ShellHacks rules review and historical observations are preserved in research_rules_2026.json and dataset.json in the parent study folder.

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Only visibly released public descriptions and Non-CEII sources were used; no secure models or reconstructed redactions. An unrelated internship listing in the archive was not analyzed. Official status statements are dated evidence, not a live feed; absence from a later report is not proof of completion. Smartsheet-derived companion template retained; Smartsheet is not the report author or sponsor.